

Technical Comments

Comment on “Novel Approach for Selecting Low-Fidelity Scale Factor in Multifidelity Metamodeling”

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I. Objective of This Comment

THE study “Novel Approach for Selecting Low-Fidelity Scale Factor in Multifidelity Metamodeling” by Shu et al. [1] proposed the approach for selecting a low-fidelity scale factor for additive scaling function-based multifidelity metamodeling to improve the modeling accuracy. To demonstrate the accuracy and efficiency of their approach, they compared it with four existing metamodeling methods, including the hierarchical kriging (HK) approach [2,3], for several example cases (one-dimensional numerical example, four multidimensional numerical examples, and an engineering case). However, while reproducing their results, it was found that the HK models in Ref. [1] show some discrepancies with the fully converged HK models due to their incomplete convergence. Therefore, this comment aims to correct their HK results by presenting newly calculated results from HK models retrained with the same dataset. The Python source code used for the retraining is available at https://github.com/sunwoong-yang/comment_MFModel.

II. HK Model in a One-Dimensional Numerical Example

First, we retrained the HK model in a one-dimensional numerical example (Sec. IV.A of Ref. [1]). This example was adapted from Zheng et al. [4]; Eq. (8) in Ref. [1] is as follows:

$$\begin{cases} f_h(x) = 0.5 \sin(4\pi \sin(x + 0.5)) + (x + 0.5)^{2/3} \\ f_l(x) = 0.5 \sin(4\pi \sin(1.1x + 0.4)) + (1.1x + 0.5)^{2/3} - 0.2 \end{cases} \quad (1)$$

However, Eq. (1) should be revised as follows (Eq. (24) in Ref. [4]):

$$\begin{cases} f_h(x) = 0.5 \sin(4\pi \sin(x + 0.5)) + \frac{(x + 0.5)^2}{3} \\ f_l(x) = 0.5 \sin(4\pi \sin(1.1x + 0.4)) + \frac{(1.1x + 0.5)^2}{3} - 0.2 \end{cases} \quad (2)$$

The high-fidelity and low-fidelity models of the revised version then matched exactly with Fig. 1 in Ref. [1], indicating that Shu et al. [1] also used Eq. (2) for their numerical test.

The identical spatial correlation function was used for the fair comparison (Eq. (2) in Ref. [1]), and none of the other parameters were arbitrarily modified or tuned from the original HK paper [2]. The fully converged HK model is shown in Fig. 1a; the incompletely

converged HK results in Ref. [1] are also shown for comparison (dashed lines). It can be verified that the poorly converged HK prediction was inaccurate with respect to the true function (solid black line, high-fidelity model) due to its overly wavy scaling function. However, the fully converged HK model shows the smooth scaling function, resulting in an accurate metamodel. Additionally, Fig. 1b compares our fully converged HK with the results of the proposed approach in Ref. [1]. Overall, they have approximately identical metamodels (their scaling functions differ because they have different forms of scaling functions, as was mentioned in Ref. [1]). The following are optimized weight parameters for our HK model: $\theta_{LF} = 13.21587962$ and $\theta_{HF} = 2.84380232$ (the same HK model can be reproduced with the same training data and weight parameters).

The accuracy metrics of the converged HK model are calculated to compare the results quantitatively, which are summarized in Table 1. We randomly sampled 200 points to estimate the max absolute error (MAE) and root-mean-square error (RMSE), as in Ref. [1]. The random process was repeated 100 times because these metrics can be sensitive to the randomness of the sampled points; therefore, RMSE and MAE in Table 1 are averaged values. We verified again that the retrained HK model shows a significant improvement compared to the HK model in Ref. [1], but similar results to the proposed approach in Ref. [1].

III. HK Models in Multidimensional Numerical Examples

Furthermore, we retrained the HK models in multidimensional numerical examples (Sec. IV.B of Ref. [1]). Accuracy metrics in total four multidimensional numerical examples (Rosenbrock, six-hump Camel-back (SC), Hartmann 6, and Borehole functions) are recalculated. However, we failed to reproduce equivalent metrics as in Table 3 of Ref. [1]. The detailed discussions about their disagreement can be found in the Appendix.

IV. HK Model in Engineering Case

Finally, we retrained the HK model for the engineering case (Sec. IV.C of Ref. [1], application to micro-aerial vehicle fuselage) with the same dataset since Shu et al. [1] provided training and validation points in their Appendix. The true and predicted responses of the retrained HK model are presented in Fig. 2 (as Fig. 11 in Ref. [1]), and its R -squared was calculated to be 0.9922. The resultant accuracy metrics are summarized in Table 2 (as Table 6 in Ref. [1]). The retrained HK model shows the error metrics that are approximately half those in Ref. [1], indicating a significant improvement. Its optimized weight parameters are listed in Table 3.

Appendix

The error metrics in Sec. IV.B of Ref. [1] and those we recalculated had significantly different scales to be compared (except Hartmann 6 function), though they were calculated exactly as demonstrated in Ref. [1]. To justify our error metrics, we conducted Latin hypercube sampling (LHS) on each input space of Rosenbrock and Borehole examples to generate 30 points (their input spaces can be found in Ref. [1]). Figure A1 illustrates the values of $R(x_i^R)$ and $B(x_i^B)$ in ascending orders, where x_i^R and x_i^B represent the i th sample points in the input spaces of Rosenbrock and Borehole example, respectively; $R(x)$ is Rosenbrock function, and $B(x)$ is Borehole function. With these sampled points, the ranges of Rosenbrock and Borehole functions were 0–3000 and 0–2.5, respectively. Therefore, the MAE and RMSE of Rosenbrock are expected to have higher scales than those of the Borehole function; however, the error metrics in Table 3 of Ref. [1]

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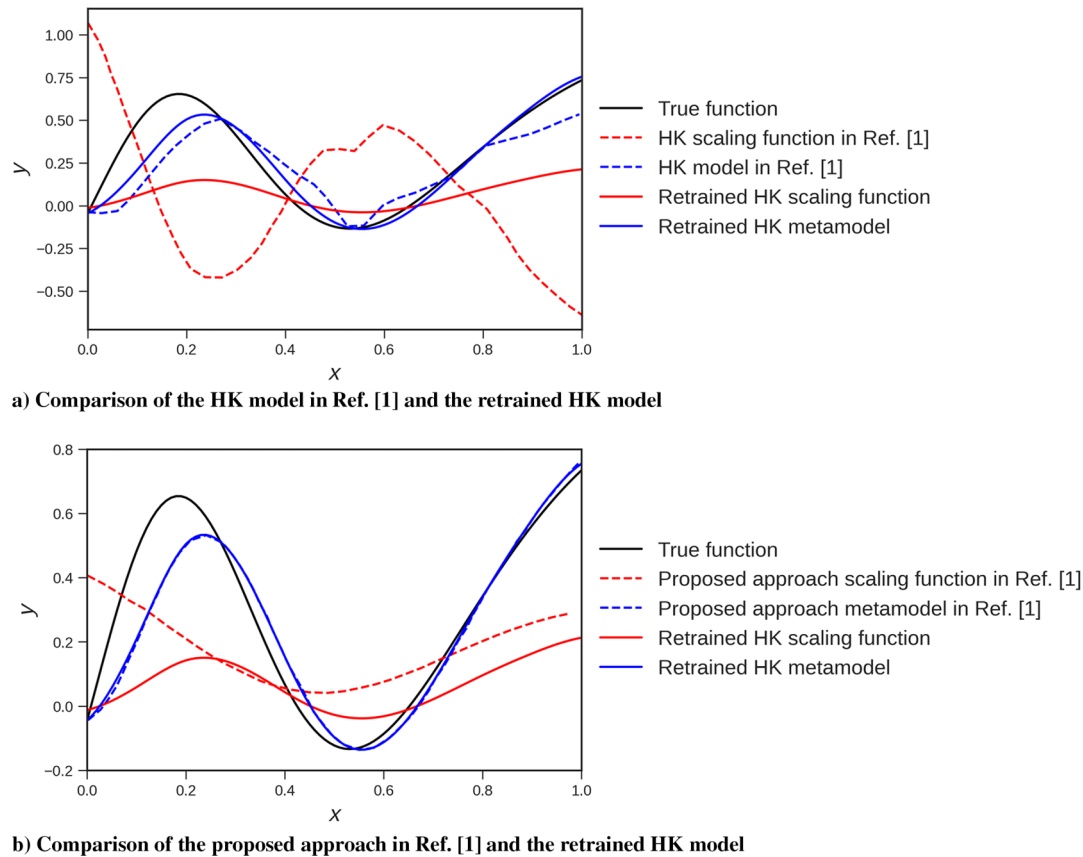


Fig. 1 Comparison of the multifidelity metamodels in Ref. [1] and the retrained HK model.

Table 1 Accuracy metrics of the retrained HK model in one-dimensional numerical example: compared with the results of Ref. [1]

Metrics	HK model of Ref. [1]	Retrained HK model	Proposed approach of Ref. [1]
ρ	1.8436	0.7180	0.7414
MAE	0.3952	0.2778	0.2792
RMSE	0.1748	0.1038	0.1058

Table 3 Optimized weight parameters of the retrained HK model (six design variables were standardized before retraining the HK model)

	A_1	A_2	A_3	D	L	t
θ_{LF}	3.04×10^{-4}	2.45×10^{-4}	6.89×10^{-4}	6.11×10^{-3}	7.01×10^{-3}	6.40×10^{-2}
θ_{HF}	1.26×10^{-4}	4.04×10^{-6}	7.64×10^{-2}	1.57×10^{-1}	1.53×10^{-1}	1.16×10^{-1}

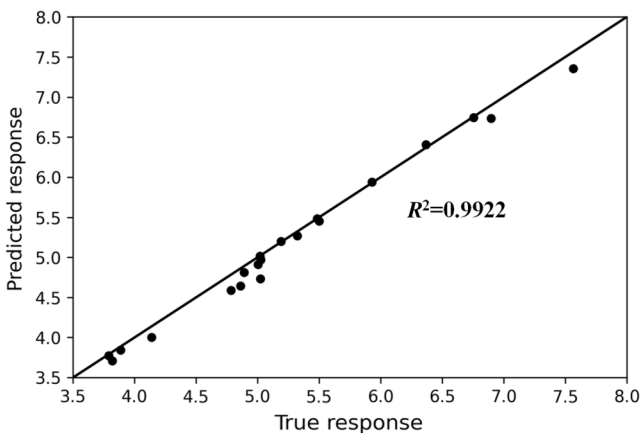


Fig. 2 True and predicted responses of the retrained HK model.

Table 2 Accuracy metrics of the retrained HK model in engineering case: compared with the results of Ref. [1]

Metrics	HK model of Ref. [1]	Retrained HK model	Proposed approach of Ref. [1]
MAE	0.6569	0.2897	0.1893
RMSE	0.2312	0.1215	0.0882

show very different scales. MAE of the HK model in the Rosenbrock case ranged from 0.1861 to 12.9070 and that in the Borehole case ranged from 801.3234 to 1530.8189. Considering the scales of their functions, we suspect that their method to calculate error metrics in these numerical examples (Table 3 in Ref. [1]) may differ from those stated in Ref. [1]. Though we failed to reproduce the results in Ref. [1], our recalculated error metrics were compared with them for clarity (see Table A1).

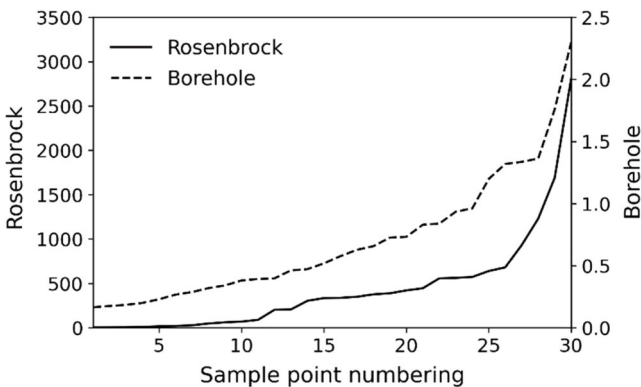


Fig. A1 Ranges of Rosenbrock and Borehole functions for 30 randomly sampled points using LHS.

Table A1 Accuracy metrics of the retrained HK models in multidimensional numerical examples: compared with the results of Ref. [1]

Function	Case	Metrics	HK model in Ref. [1]	Retrained HK model
Rosenbrock	1	MAE	12.9070	439.2971
		RMSE	2.1739	67.8673
	2	MAE	0.6091	178.7216
		RMSE	0.0923	30.7339
	3	MAE	0.5020	87.5640
		RMSE	0.0741	13.6409
	4	MAE	0.1861	61.2612
		RMSE	0.1841	8.1192
SC	1	MAE	0.0284	5.7011
		RMSE	0.0054	1.1338
	2	MAE	0.0218	5.4246
		RMSE	0.0038	1.0525
	3	MAE	0.0014	2.5447
		RMSE	0.0002	0.5865
	4	MAE	0.0014	2.3772
		RMSE	0.0002	0.5313
Hartmann 6	1	MAE	0.9576	1.0154
		RMSE	0.1847	0.2567
	2	MAE	0.8846	0.9844
		RMSE	0.1978	0.2356
	3	MAE	0.8072	0.8885
		RMSE	0.1695	0.2275
	4	MAE	0.8491	0.8888
		RMSE	0.1764	0.2081
Borehole	1	MAE	1530.8189	1.9995
		RMSE	349.5859	0.5079
	2	MAE	1199.0202	1.9152
		RMSE	321.2086	0.4753
	3	MAE	1037.4457	1.4582
		RMSE	233.8206	0.3646
	4	MAE	801.3234	1.4021
		RMSE	190.7707	0.3377

Acknowledgments

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